Comelia Soltanirad

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EDUCATION

Case Western Reserve University, School of Medicine, Cleveland, OH
Master of Science in Epidemiology and Biostatistics — Graduation: August 2026

The Ohio State University, College of Arts & Sciences, Columbus, OH
Bachelor of Arts in Chemistry — Graduation: May 2025

PUBLICATIONS

Clark, Ted M., et al. "Comparing the performance of college chemistry students with ChatGPT for calculations involving acids and bases." *Journal of Chemical Education.*

PRESENTATIONS

Comparing the Performance of Students with ChatGPT for Calculations Involving Acids and Bases. Ohio PKAL Regional Spring Meeting, Kenyon College, Gambier, Ohio. 5/13/2023.

A Literature Review of Student Sense of Belonging in STEM Courses. Ohio PKAL Regional Spring Meeting, Otterbein University, Westerville, Ohio. 4/6/2024.

RESEARCH EXPERIENCE

Xi Lab (Northwestern University, Feinberg College of Medicine) - Research Assistant
February 2026 - July 2026

- Analyzed nationally representative Youth Risk Behavior Surveillance System (YRBS) data to identify patterns in substance use and suicide-related behaviors among U.S. high school students, highlighting groups with higher levels of risk
- Developed and evaluated logistic regression models across 11 substance-use variables, generating risk estimates to compare suicide-related outcomes
- Produced statistical tables and visualizations to communicate model findings, identify patterns in suicide-related risk, and inform adolescent mental health resources

Clark Lab (Ohio State University, Department of Chemistry and Biochemistry) - Student Research Assistant
April 2023 – May 2025

- Assessed LLM reliability by comparing AI-generated outputs with human responses and expected solutions across 10 undergraduate chemistry problems
- Organized model responses into comparison tables via Excel and classified recurring calculation and reasoning errors to evaluate model accuracy
- Documented error patterns and high-confidence incorrect outputs to identify model limitations and potential risks of relying on AI-generated results in an academic setting

Mihaylova Lab (Ohio State University, Comprehensive Cancer Center) - Student Research Assistant
October 2021 – May 2023

- Harvested cells for RNA and protein extraction for RT-qPCR and western blot analysis.
- Generated overexpression vectors using molecular cloning methods.
- Performed immunofluorescent and immunohistochemical staining on mouse tissues.
- Maintained mammalian tissue culture lines and assisted with routine genotyping.

TEACHING EXPERIENCE

School of Engineering, Case Western Reserve University - Teaching Assistant
August 2025 – December 2025

- Evaluated student work against established rubrics to identify errors and communicate feedback to improve student learning

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Field Experience
Mettler Toledo - Lab Tech Support Intern
May 2023 – August 2023

- Executed customer analytical methods, including Karl Fischer titration and thermal analysis (DSC, TGA), to evaluate instrument performance
- Applied critical thinking and attention to detail to identify measurement issues, troubleshoot results, and communicate findings to customers and team members

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WORK EXPERIENCE
Betts at The Schofield Hotel (Kimpton Hotels) — Cleveland, OH
Host, Server | Aug 2025 - September 2026

- Provide elevated, attentive service in a high-end hotel dining environment
- Greet and seat guests, manage reservations, walk-ins, and waitlists to optimize flow and table turnover
- Serve food and beverages, ensuring accuracy, timing, and high-quality guest experience
- Collaborate with servers, bartenders, and management to maintain seamless operations
- Deliver knowledgeable recommendations on menu items, cocktails, and dietary accommodations

Society Lounge — Cleveland, OH
Server | Jan 2026 - September 2026

- Delivered attentive service across multiple tables at once in a fast-paced environment
- Managed multiple tables simultaneously while maintaining accuracy in orders and timing
- Collaborated with bar to ensure accurate drink service
- Handled POS transactions, guest requests, and issue resolution

Katalina's Café — Columbus, OH
Server | May 2025–Aug 2025

- Delivered service at an award-winning restaurant featured on Diners, Drive-Ins, and Dives
- Processed payments, coordinated with kitchen staff, and maintained accuracy and pace during peak service periods

Personal Trainer | Recreation & Physical Activity Center (RPAC), The Ohio State University, Columbus, OH
January 2024 – August 2025

- Conducted fitness assessments to understand clients' fitness levels and goals.
- Designed and implemented individualized workout programs tailored to client needs.
- Provided guidance on nutrition, lifestyle, and exercise techniques for optimal outcomes.

Barista | Kittie's Cakes, Columbus, OH
June 2022 - November 2024

- Mastered espresso preparation and fulfilled 100+ daily customer orders.
- Demonstrated comprehensive menu knowledge and offered personalized drink recommendations. • Collaborated with baristas, staff, and managers to streamline operations.

Barista | Rohr Café at The Ohio State University, Columbus, OH
August 20230 - May 2024

- Operated espresso machines, blenders, and beverage equipment with precision.
- Maintained cleanliness and organization of the café, including dining and prep areas.

CAMPUS INVOLVEMENT

Director, Sustainability Committee, Undergraduate Student Government, The Ohio State University

May 2024 - May 2025
• Engaged faculty to organize sustainability-focused events across campus.
• Managed a team of 6 deputy directors and 20 committee representatives.

Deputy Director, Sustainability Committee, Undergraduate Student Government, The Ohio State University

May 2022 - May 2024
• Collaborated with six local businesses to host OSU's first Sustainable Fashion Show.
• Delivered lectures on sustainable fashion and conscious consumption.
• Organized two clothing exchanges to promote slow fashion.
• Planned a sustainable yoga class for 35 students to advocate sustainable wellness.

Committee Representative, Sustainability Committee, Undergraduate Student Government, The Ohio State University

September 2021 - May 2022
• Organized a campus-wide clothing exchange, coordinating collection, sorting, and outreach. • Designed surveys to gather student feedback about dormitory sustainability and living conditions.

Director of Finance | Scarlette Magazine, The Ohio State University
May 2023 - May 2024
• Developed semester budgets and optimized resource allocation.
• Aligned organizational finances with executive board goals and initiatives.

SKILLS

● **Programming & Data Science:** Python (pandas, NumPy, SciPy, scikit-learn), R, SAS, Git/GitHub
● **Statistical Methods:** Regression modeling, ANOVA, Bayesian methods, Monte Carlo simulation, time series analysis, resampling (bootstrap), model evaluation & diagnostics
● **Machine Learning:** Supervised/unsupervised learning, feature selection (LASSO/Ridge), random forests, gradient boosting, survival analysis (Cox models, Kaplan–Meier), predictive modeling workflows
● **Risk & Survival Analysis:** Time-to-event analysis, Cox proportional hazards models, parametric survival models, accelerated failure time (AFT) models, competing risks, censoring (right/left/truncation)
● **Data Handling:** Data cleaning, transformation, normalization, and preparation of analysis-ready datasets
● **Visualization & Reporting:** ggplot2, tidyverse, Quarto/RMarkdown for reproducible reporting
● **Laboratory Techniques:** RT-qPCR, Western Blotting, Molecular Cloning, Tissue Culture, Immunofluorescence, Immunohistochemistry, DSC, TGA
● **Other Skills:** Scientific writing, data communication, teamwork, time management

CERTIFICATIONS

ACE Certified Personal Trainer (CPT) – American Council on Exercise (10/2023–Present)
CPR Certification – American Red Cross, Columbus, Ohio (11/2025)